

Fengping (Helen) Hu

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OVERVIEW

Vision scientist and UX researcher with 8 years of experience in behavioral and neuroscience research. Expertise in spatial vision (limits of visual perception, reading, face/object recognition). Translated vision science into HCI applications such as immersive video-viewing. Skilled in designing user studies and modeling user experience. Proficient in communicating complex findings to technical and non-technical audiences.

EDUCATION

Ph.D. in Cognition & Perception, New York University

New York, NY

- Advisors: Dr. Jonathan Winawer, Dr. Denis Pelli; GPA: 4.00/4.00

Expected Jun. 2028

B.A. in Cognitive Science, Carleton College

Northfield, MN

- Minors: neuroscience, music performance; GPA: 3.99/4.00; *summa cum laude*

Sept. 2019 – Jun. 2023

EXPERIENCE

PhD Research Intern

Sunnyvale, CA

Dolby Laboratories, Inc.

Jun. 2026 – Sept. 2026

- Designed, programmed, and conducted an end-to-end user study about how subtitle design affects viewing experience in high-dynamic-range (HDR) video (paper in prep).
- Built a video-stimulus generation pipeline using speaker detection and image analysis to systematically manipulate viewing conditions in HDR content.
- Optimized text placement in videos, maximizing readability, comfort, and engagement.

Doctoral Researcher: Limits of vision; online psychophysics

New York, NY

Winawer Lab & Pelli Lab, New York University

Sept. 2023 – present

- Designed and validated the first online gaze-control method that eliminates gaze deviations from fixation. Leveraged eye-tracking and qualitative interviews to optimize gaze accuracy and user experience (first-author publication, *Journal of Vision*; first-author poster, *VSS 2025*). Scaled the method to collect 7 online datasets of visual performance and reading speed ($N > 800$, first-author poster, *VSS 2026*).
- Analyzed large fMRI datasets, using statistical modeling to find neural correlates that explain individual differences in the limits of visual perception and reading speed.
- Developed a new account of how visual crowding limits reading speed; used mathematical modeling to support the proposed mechanism, challenging the existing dominant theory of reading (paper in prep).
- Partnered with software engineers to test and refine a web app for online psychophysics; translated research requirements into actionable product feedback and validated software releases through behavioral testing.

Research Intern: Symmetry perception; visual deprivation and low vision

Cambridge, MA

Sinha Lab, Massachusetts Institute of Technology

Jun. 2022 – Sept. 2022

- Led an online psychophysical study ($N = 40$) on the perception of symmetrical patterns across wide separations (first-author publication, *Vision Research*).
- Analyzed a 12-year longitudinal dataset ($N > 100$) on the development of face and object recognition in children who experienced visual deprivation and low vision (co-first author publication, *Scientific Reports*).
- Designed and ran experiments for translating a user's viewing distance to pixel-level spatial resolution.

Research Intern: Audiovisual integration; listening effort

Northfield, MN

Perception Lab, Carleton College

Sept. 2019 – Jun. 2023

- Executed experiments on audiovisual integration and listening effort; pre-processed and analyzed speech data (co-author poster, *Psychonomic Society 2022*).

SKILLS

- Programming:** Python, MATLAB, Swift, Shell, R | **Tools:** Git/GitHub, Prolific, Adobe Creative Cloud
- Statistical & Computational Modeling:** Advanced regression, multivariate statistical analysis (e.g., linear and nonlinear mixed-effects modeling), Bayesian statistics, model comparison
- Specialized Methods:** human perceptual testing (in person and online), eye tracking, fMRI, EEG, MEG